

Human Anatomy and Neuroanatomy Principles

Anatomy — MBBS 1st Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 1st Year
Medical Subject	Anatomy
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
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Chapter 1: Musculoskeletal and Peripheral Nervous System

Brachial Plexus Organization and Neurological Lesions (Erb's and Klumpke's Palsy)

The brachial plexus provides somatic nerve supply to the upper limb. It is formed by the anterior rami of cervical spinal nerves C5, C6, C7, C8, and T1. The plexus is organized anatomically from proximal to distal into five roots, three trunks (superior, middle, inferior), six divisions (three anterior, three posterior), three cords (lateral, posterior, medial named relative to the axillary artery), and terminal branches.

The superior trunk is formed by the union of C5 and C6 roots. An injury to Erb's point (the junction of C5 and C6) typically occurs during difficult vertex deliveries with excessive traction on the head or falls onto the shoulder. This produces Erb-Duchenne palsy, characterized clinically by the 'waiter's tip' or 'porter's tip' deformity. In this deformity, the arm hangs by the side in adduction and internal rotation, the elbow is extended, and the forearm is pronated.

Conversely, Klumpke's palsy involves avulsion of the inferior trunk (C8 and T1 nerve roots). This classic lesion arises from upward traction on the arm, such as when a person grabs a tree branch while falling or during breech deliveries. Paralysis affects intrinsic hand muscles supplied by T1, resulting in a claw hand accompanied by ipsilateral Horner syndrome due to interruption of sympathetic fibers.

Clinical Pearl: Erb's palsy affects C5-C6 (waiter's tip posture; loss of shoulder abductors and lateral rotators). Klumpke's affects C8-T1 (claw hand deformity and ipsilateral Horner syndrome).

	Clinical Posture	Primary Mechanism
	Adducted, internally rotated shoulder, extended elbow, pronated forearm ('Waiter's tip')	Shoulder dystocia
	Intrinsic hand muscle paralysis, claw hand, loss of sensation on medial hand/forearm	Breech delivery

Chapter 1: Musculoskeletal and Peripheral Nervous System

Carpal Tunnel Anatomy and Median Nerve Compression Syndrome

The carpal tunnel is an osteofibrous canal located on the anterior aspect of the wrist. Its posterior and lateral boundaries are formed by the deep concave arch of carpal bones (scaphoid, trapezium, pisiform, and hook of hamate). Its anterior boundary is formed by the thick fibrous flexor retinaculum (transverse carpal ligament).

Ten structures pass through the carpal tunnel: four tendons of flexor digitorum superficialis, four tendons of flexor digitorum profundus, one tendon of flexor pollicis longus, and a single neural structure—the median nerve. The median nerve lies superficial to the flexor tendons directly beneath the flexor retinaculum.

Compression of the median nerve produces Carpal Tunnel Syndrome (CTS). Patients present with paresthesia, burning pain, and numbness in the lateral three and a half digits (thumb, index, middle, and radial half of ring finger). Sensation over the thenar eminence is spared because the palmar cutaneous branch branches proximal to the flexor retinaculum and passes superficial to it.

***Clinical Pearl:** In Carpal Tunnel Syndrome, thenar sensation is preserved because the palmar cutaneous branch passes superficial to the flexor retinaculum.*

Feature	Carpal Tunnel Syndrome Details
Boundary	Carpal bones posteriorly; Flexor retinaculum anteriorly
Contents	9 tendons (4 FDS, 4 FDP, 1 FPL) + Median Nerve
Provocative Tests	Phalen test (wrist flexion 60s) and Tinel sign (percussion over retinaculum)
Motor Deficit	Atrophy of thenar muscles (Abductor pollicis brevis, Flexor pollicis brevis, Opponens pollicis)

Chapter 2: Thorax, Cardiovascular and Neuroanatomy

Coronary Arterial Circulation, Territories and Cardiac Dominance

The heart receives arterial blood supply through the right and left coronary arteries, arising from ascending aorta sinuses of Valsalva. The left coronary artery (LCA) bifurcates into the Left Anterior Descending (LAD) artery and the Left Circumflex (LCx) artery. The LAD supplies the anterior two-thirds of the interventricular septum, anterior LV wall, and cardiac apex ('widow-maker' artery).

Cardiac dominance is determined anatomically by which coronary artery gives off the Posterior Descending Artery (PDA). In 85% of individuals, the PDA arises from the Right Coronary Artery (RCA), termed right-dominant circulation. In 8-10%, the PDA arises from LCx (left-dominant). In the remaining patients, both contribute (codominant).

The RCA supplies the Sinoatrial (SA) node in ~60% and the Atrioventricular (AV) node in ~90% of individuals. Consequently, acute inferior wall myocardial infarctions caused by acute RCA occlusion frequently precipitate sinus bradycardia, sinus arrest, and high-degree AV block.

Clinical Pearl: Right dominance accounts for ~85% of individuals; RCA supplies the AV node via the AV nodal artery, making inferior wall MI frequently complicated by AV nodal conduction block.

	Territory Supplied	Common Clinical Correlation
	Anterior LV wall, anterior 2/3 of septum, apex	Anterior STEMI
	Right ventricle, inferior LV wall, posterior 1/3 septum, SA/AV nodes	Inferior STEMI
	Lateral and posterior walls of left ventricle	Lateral STEMI

Chapter 2: Thorax, Cardiovascular and Neuroanatomy

Cranial Nerves and Brainstem Stroke Localization Syndromes

Twelve pairs of cranial nerves emerge from the brain. Localization within the brainstem follows the 'rule of 4s': CN I and II emerge from the forebrain; CN III and IV exit the midbrain; CN V, VI, VII, and VIII emerge from the pons or pontomedullary junction; and CN IX, X, XI, and XII exit the medulla oblongata.

Cranial nerve III (Oculomotor nerve) innervates ocular muscles and provides parasympathetic pupillary constriction via Edinger-Westphal nucleus. In uncal herniation, the temporal uncus compresses CN III against the petroclinoid ligament, causing an ipsilateral 'blown pupil' followed by 'down and out' globe displacement.

Cranial nerve VII (Facial nerve) controls facial expression. In upper motor neuron (UMN) lesions (stroke), the forehead is spared due to bilateral cortical innervation. In lower motor neuron (LMN) lesions (Bell's palsy), both upper and lower face are paralyzed on the ipsilateral side, impairing eye closure and forehead wrinkling.

Clinical Pearl: Forehead sparing distinguishes UMN facial weakness (cortical stroke) from LMN facial paralysis (Bell's palsy), where the entire ipsilateral hemiface is involved.

	Blink Reflex / Eye Closure	Etiology Exam
	Intact or minimally affected	Contralateral M
	Impaired (inability to close eye, Bell's phenomenon)	HSV-1 reactivation

Chapter 2: Thorax, Cardiovascular and Neuroanatomy

Meninges, Subarachnoid Space, CSF Circulation, and Lumbar Puncture Anatomy

The central nervous system is enveloped by three meningeal layers: the outer tough Dura Mater (pachymeninx), the middle delicate Arachnoid Mater, and the inner vascular Pia Mater intimately adhering to the brain and spinal cord parenchyma. The Pia and Arachnoid together constitute the Leptomeninges, which become acutely inflamed in meningitis.

The Subarachnoid Space lies between the arachnoid and pia mater and contains approximately 150 mL of Cerebrospinal Fluid (CSF). The spinal cord terminates at the conus medullaris at the L1-L2 vertebral body level in adults (L3 in neonates). Below this level, the subarachnoid space expands into the Lumbar Cistern containing the cauda equina, extending to the second sacral vertebra (S2).

Lumbar puncture (LP) is performed at the L3-L4 or L4-L5 intervertebral interspace (identified by Tuffier's line connecting the highest points of the iliac crests). The spinal needle penetrates sequentially: skin, subcutaneous tissue, supraspinous ligament, interspinous ligament, ligamentum flavum ('pop' sensation), epidural space, dura mater, and arachnoid mater into the subarachnoid space.

Clinical Pearl: Lumbar puncture enters the subarachnoid space at L3-L4 or L4-L5, below the conus medullaris (L1-L2), safely traversing the ligamentum flavum and dura-arachnoid without injuring the spinal cord.

Anatomical Characteristic	Clinical Significance
Fibrous longitudinal midline ligaments	Provides resistance and needle guidance
Thick, elastic yellow ligament connecting laminae	Distinct tactile 'pop' felt upon needle penetration
Tough fibrous sheath enclosing subarachnoid space	Second tactile 'give'; CSF flows upon stylet withdrawal
CSF-filled cavity containing cauda equina roots	Target site for opening pressure and diagnostic CSF collection